

Performance evaluation of multi-objective techniques for ZDT Function: A comparative study



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Performance Evaluation of Multi-objective techniques for ZDT Function: A Comparative Study

Abstract

In Multi-objective Optimization Problems (MOPs), there is a need for multiple objective optimizations such that more than one conflicting objective must be minimized at once, hence obtaining a set of optimal solution points referred to as Pareto Front. Due to difficulty of exact approaches, evolutionary algorithms are popularly used. In this paper, comparison between NSGA-II and SPEA-II which are the most common multi-objective evolutionary algorithms is carried out using four ZDT benchmarks: ZDT1 (convex function), ZDT2 (concave function), ZDT3 (discontinuous function) and ZDT4 (multi modal function). Both algorithms were executed with the same parameters; population and archive sizes (100 individuals each), crossover and mutation operators (SBX, polynomial), generation limit (200).

Evaluation was based on convergence (size of the non-dominated front per generation), quality of final front, IGD, and HV. The results indicated that SPEA-II converges faster in convex (ZDT1) and concave (ZDT2) functions and gets the completely non-dominated front in 100-150 generations compared to NSGA-II in 150-200 generations.

But in contrast, in case of the discontinuous problem of ZDT3, NSGA-II reaches stability earlier, between generation 100 and 150, while SPEA-II shows temporary drops in performance. In the highly multi-modal ZDT4 test problem, both algorithms manage to achieve the complete set of non-dominated solutions, but NSGA-II remains monotone, while SPEA-II experiences occasional drops in performance due to poor archive quality until reaching stability. Generally, SPEA-II produces slightly better IGD and HV values compared to NSGA-II on all test problems. Nevertheless, NSGA-II demonstrates superior stability when dealing with discontinuity and multi-modality. Thus, it appears that SPEA-II performs better on problems with smooth Pareto front, while NSGA-II is more reliable for other test problems.

1. Introduction

Many real-world problems involve the need to optimize several objectives that conflict with each other. For instance, increasing the fuel efficiency of cars might lead to reduced performance or compromised safety. As opposed to a single optimal solution, such problems produce a collection of optimal solutions called the Pareto optimal set. Multi-objective optimization problems (MOPs) refer to such problems.

The weighted sum method and epsilon-constraint methods are some common traditional techniques that perform adequately in simpler problems but lack efficiency in larger and more complicated problems. Meta-heuristic techniques overcome this drawback by offering an approximation of the Pareto front in an acceptable period.

There are many MOEAs used in solving multi-objective optimization problems. However, the most common ones are NSGA-II and SPEA-II. Both algorithms employ non-dominated sorting and preserve diversity in their operations. The only difference between the two is their fitness function and elitism strategy.

This report introduces a comprehensive comparison between NSGA-II and SPEA-II algorithms based on four benchmarks known as ZDT1, ZDT2, ZDT3, and ZDT4 (Zitzler, Deb & Thiele, 2000). The above-mentioned benchmarks differ from one another in terms of their Pareto front features that are convex, non-convex, discontinuous, and multi-modal. We will assess the algorithms using IGD, HV, and spread indicators.

The two pioneering MOEAs to be considered in this report are:

- NSGA-II (Non-dominated Sorting Genetic Algorithm II): Introduced by Deb et al. (2002); well known for its fast non-dominated sorting algorithm and crowding distance method for maintaining diversity in $O(MN^2)$.
- SPEA-II (Strength Pareto Evolutionary Algorithm 2): Introduced by Zitzler et al. (2001); known for its use of an external archive, strength-based selection procedure, and k-nearest neighbour density estimation.

2. Problem Definition: ZDT1

2.1 Zitzler-Deb-Thiele (ZDT) Test Suite

The ZDT benchmark suite is a series of multi-objective optimization test problems designed systematically by Zitzler, Deb, and Thiele. Each ZDT problem is created in such a way that the Pareto optimal frontier demands different convergence and diversity characteristics from the solver. The general form of the ZDT problem is:

$$\text{Minimize } f_1(X),$$

$$\text{Minimize } f_2(X) = g(X)h(f_1(X), g(X)),$$

$$0 \leq x_i \leq 1, i = 1, \dots, n.$$

where f_1 depends only on the first decision variable x_1 , g captures the influence of the

remaining variables, and h defines the shape of the Pareto-optimal front.

2.2 ZDT1 Formulation

ZDT1 is a 30-variable ($n = 30$) bi-objective minimization problem defined as follows:

$$\begin{aligned} \text{Minimize } f_1(X) &= x_1, \\ \text{Minimize } f_2(X) &= g(X)h(f_1(X), g(X)), \\ g(X) &= 1 + \frac{9}{n-1} \sum_{i=2}^n x_i, \\ h(f_1, g) &= 1 - \sqrt{f_1/g}. \end{aligned}$$

Key properties of ZDT1:

- It is an $n = 30$ -variable problem, meaning the decision space is 30-dimensional.
- The Pareto-optimal front is convex, defined by $g(x) = 1$ (all $x_i = 0$ for $i \geq 2$) and $f_2 = 1 - \sqrt{f_1}$.
- The objective space exhibits a strong non-linear trade-off between f_1 and f_2 .
- The problem is designed to test convergence towards the Pareto-optimal front and uniform distribution along it.
- The true Pareto front lies in the region $f_1 \in [0, 1]$, $f_2 = 1 - \sqrt{f_1}$, requiring algorithms to drive $x_i \rightarrow 0$ for $i \geq 2$.

Apart from ZDT1, we have used more three ZDT problems with n decision variables, $x_i \in [0,1]$, two objectives to minimize, and the following definitions.

➤ ZDT1 (Convex)

$$\begin{aligned} \text{Minimize } f_1(X) &= x_1, \\ \text{Minimize } f_2(X) &= g(X)h(f_1(X), g(X)), \\ g(X) &= 1 + \frac{9}{n-1} \sum_{i=2}^n x_i, \\ h(f_1, g) &= 1 - \sqrt{(f_1/g)}. \end{aligned}$$

- It is a $n = 30$ variable problem.
- It has a convex Pareto-optimal front.

➤ ZDT2 (Non-convex)

$$\begin{aligned} \text{Minimize } f_1(X) &= x_1, \\ \text{Minimize } f_2(X) &= g(X)h(f_1(X), g(X)), \\ g(X) &= 1 + \frac{9}{n-1} \sum_{i=2}^n x_i, \\ h(f_1, g) &= 1 - (f_1/g)^2. \end{aligned}$$

- It is a $n = 30$ -variable problem.
- It has a nonconvex Pareto-optimal front.

➤ ZDT3 (Discontinuous)

$$\begin{aligned} \text{Minimize } f_1(X) &= x_1, \\ \text{Minimize } f_2(X) &= g(X)h(f_1(X), g(X)), \end{aligned}$$

$$g(X) = 1 + \frac{9}{n-1} \sum_{i=2}^n x_i,$$

$$h(f_1, g) = 1 - \sqrt{\left(\frac{f_1}{g}\right) - \left(\frac{f_1}{g}\right) \sin(10\pi f_1)}.$$

- It is a $n = 30$ -variable problem.
- It has a number of disconnected Pareto-optimal fronts.

➤ ZDT4 (Multi-modal)

$$\text{Minimize } f_1(X) = x_1,$$

$$\text{Minimize } f_2(X) = g(X)h(f_1(X), g(X)),$$

$$g(X) = 1 + 10(n-1) + \sum_{i=2}^n (x_i^2 - 10 \cos(4\pi x_i)),$$

$$h(f_1, g) = 1 - \sqrt{(f_1/g)}.$$

- It is a $n = 10$ -variable problem. Except for x_1 , $-5 \leq x_i \leq 5$.
- It has a convex Pareto-optimal front. There exists 21^9 solutions.

All problems have to be minimized for both f_1 and f_2 .

3. Comparative Study of Metaheuristic Techniques

3.1 NSGA-II (Non-Dominated Sorting Genetic Algorithm II)

NSGA-II is elitist multi-objective evolutionary algorithms that evolve a constant number of population through several generations based on non-domination sort and crowding distance as the selection mechanisms.

- **Initialisation:** A parent population P_0 consisting of $N = 100$ random solutions is created and evaluated on all the objectives.
- **Generation of Offspring:** In each generation t , a child population Q_t is generated with N random solutions using binary tournament selection, simulated binary crossover with probability 0.9 and $\eta_c = 15$, and polynomial mutation with probability 0.1 and $\eta_m = 20$.
- **Population Combination and Partitioning into Non-Domination Fronts:** Parent and offspring populations are first combined to form a population of size $2N$, represented by $R_t = P_t \cup Q_t$, and this population is divided into non-domination fronts $F_1, F_2, F_3, \dots$ through fast non-domination sorting, with crowding distances calculated for individuals on each front.
- **Parent Selection Process:** For forming the new population P_{t+1} , fronts F_1 and onward are taken as a whole, and if at any stage the population exceeds N individuals, remaining individuals are selected in descending crowding distance order from that front.
- **Termination:** This cycle repeats itself for 200 iterations. The non-dominated set of solutions from the last iteration makes up the approximate Pareto optimal

frontier. Unlike SPEA2, NSGA-II does not use any external archive; the concept of elitism is taken care of directly by incorporating P_t into R_t .

3.2 SPEA-II (Strength Pareto Evolutionary Algorithm II)

The SPEA-II algorithm is an elitist MOEA which utilizes a population as well as an archive of predefined size to save and evolve the most promising solutions over time.

- **Initialisation:** An initial population P_0 is created randomly with the size $N = 100$ and also an empty archive A_0 . The two are evaluated using all objective functions.
- **Assignment on Fitness:** For every individual belonging to the collective population $A_t \cup P_t$ in every generation, the fitness value is calculated. Here, the strength $S(i)$ of an individual is the total number of solutions dominated by that individual. The raw fitness $R(i)$ is thus the sum of the strengths of all individuals dominating the individual in consideration. The density $D(i)$ is calculated based on the distance from the k -th closest individual, where $k = \sqrt{(N + \tilde{N})} = 14$.
- **Environmental Selection:** All individuals where $F(i) < 1$ are transferred to the next archive A_{t+1} . If the archive contains more than N individuals, the truncation operation is applied and the individual with the minimum distance to its nearest neighbour is eliminated until the archive reaches size N . Otherwise, individuals that are dominated and have the least fitness value are added to the archive.
- **Reproduction Selection and Variation:** The new population P_{t+1} is formed using binary tournament selection on the archive, SBX crossover ($0.9, \eta_c = 15$), and polynomial mutation ($0.1, \eta_m = 20$).
- **Stopping Condition:** The process continues for 200 iterations. After completion, the non-dominated solutions found in the last archive constitute the approximate Pareto-optimal front.

4. Experimental Setup

Parameter	NSGA-II	SPEA-II
Population size	100	100
Archive size	-	100
Generations	200	200
Crossover probability	0.9	0.9
Mutation probability	0.1	0.1
SBX distribution index	15	15
Polynomial mutation index	20	20

Performance Metrics:

There are three commonly applied measures for evaluating and comparing the quality of approximated Pareto fronts generated by the NSGA-II and SPEA-II algorithms.

- **Inverted Generational Distance (IGD):** IGD calculates the mean Euclidean distance between each point from the true Pareto front and its nearest member in the approximated front. This criterion takes into account both convergence and diversity, with smaller values being preferable.
- **Hypervolume (HV):** HV is the volume of the objective space region covered by the approximated front, considering a predefined reference point. Both convergence and diversity

- are taken into consideration, and higher values are preferred.
- **Spread (Δ):** Δ -Spread measures the distribution and spread of solutions on the identified Pareto front. A low value of Δ -Spread indicates good coverage and distribution of solutions along the Pareto front.

Due to the availability of analytical formulas for obtaining Pareto front for ZDT1, ZDT2, ZDT3, and ZDT4 problems, all the above-mentioned performance metrics can be precisely calculated using the reference Pareto front.

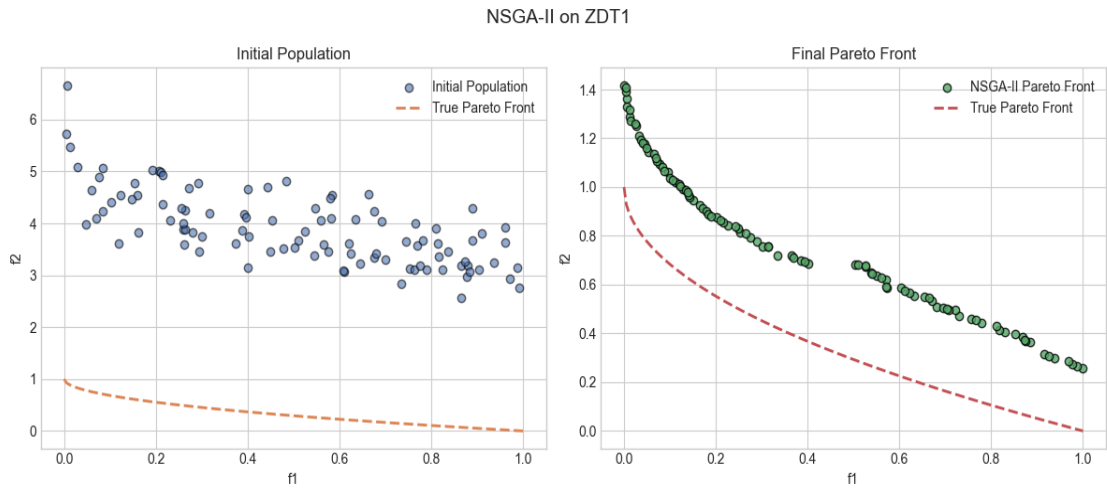
5. Results and Discussion

5.1 NSGA-II Performance

- **ZDT1 (Convex Pareto Front)**

Generations : 200

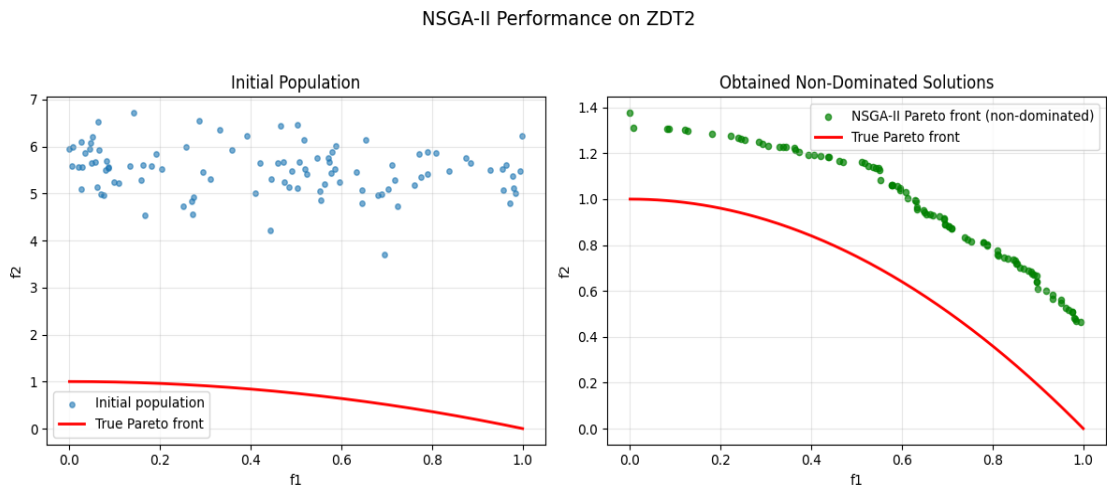
Front-0 size at the end : 90



- **ZDT2 (Non-convex, concave front)**

Generations : 200

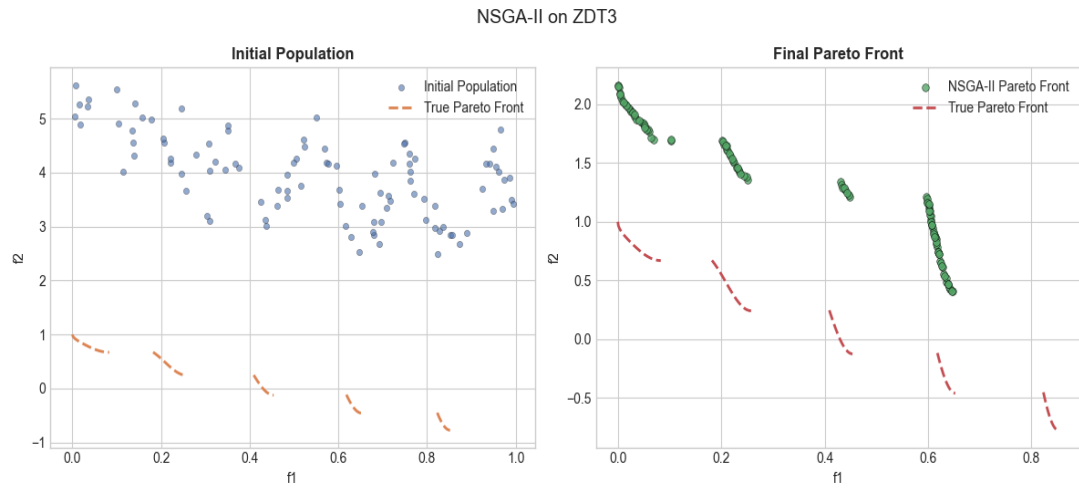
Front-0 size at end : 70



- **ZDT3 (Discontinuous front)**

Generations : 200

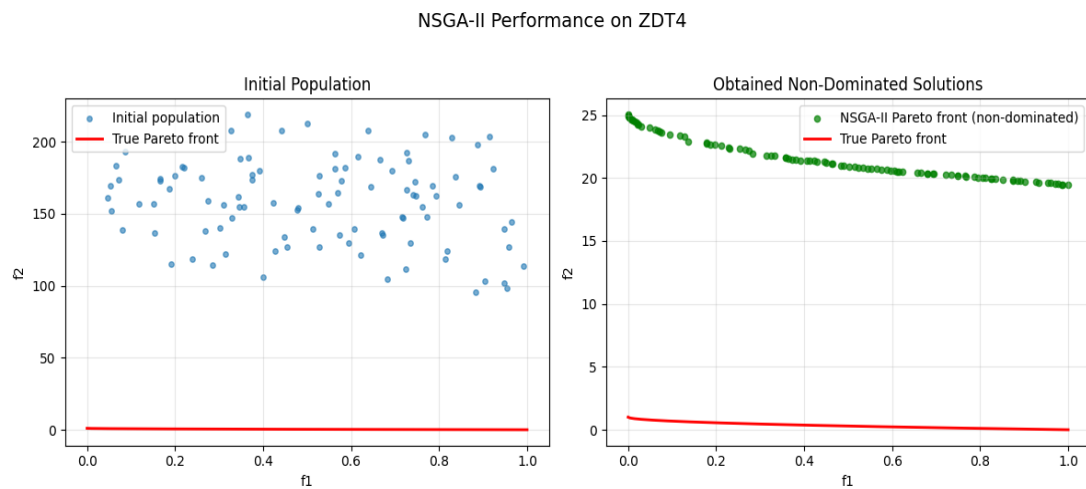
Front-0 size at end: 80



- **ZDT4 (Multi-modal)**

Generations : 200

Front-0 size at end : 95



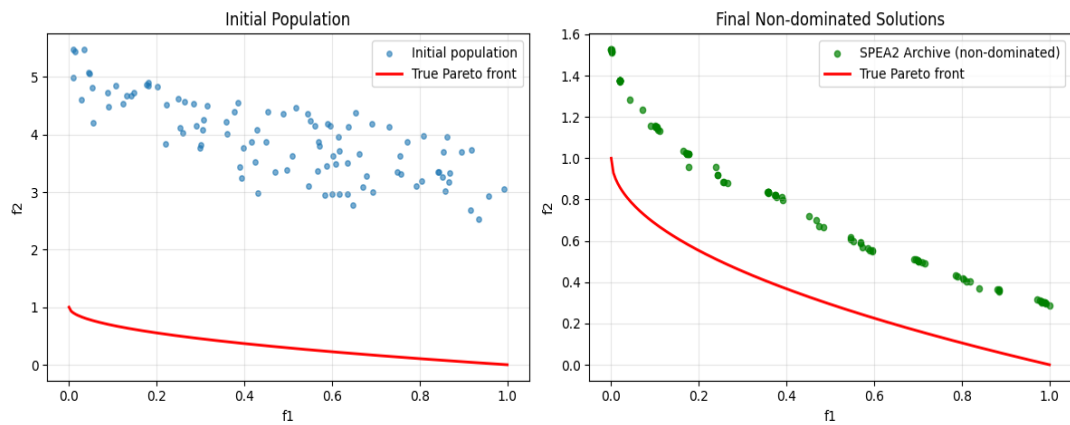
5.2 SPEA-II Performance

- **ZDT1 (Convex)**

Generations: 200

Front-0 size at end: 50

SPEA2 Performance on ZDT1

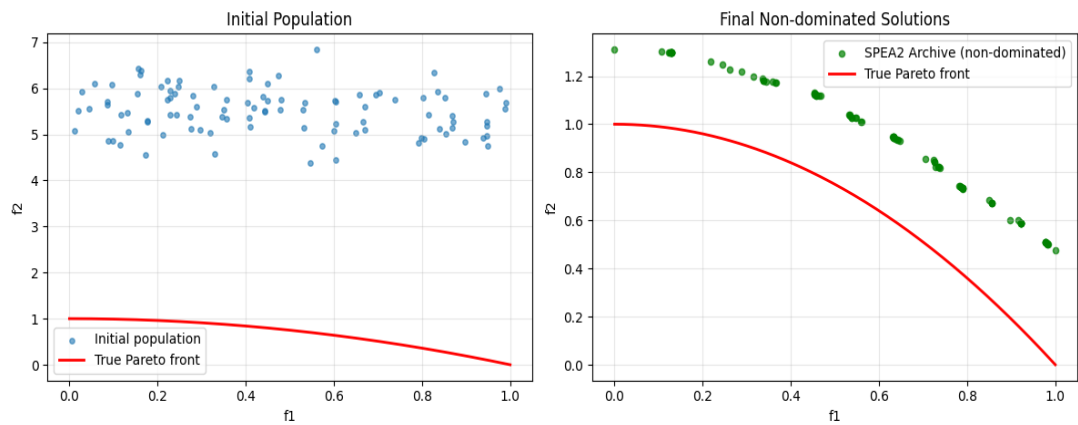


- **ZDT2 (non-convex/concave)**

Generations: 200

Front-0 size at end: 70

SPEA2 Performance on ZDT2

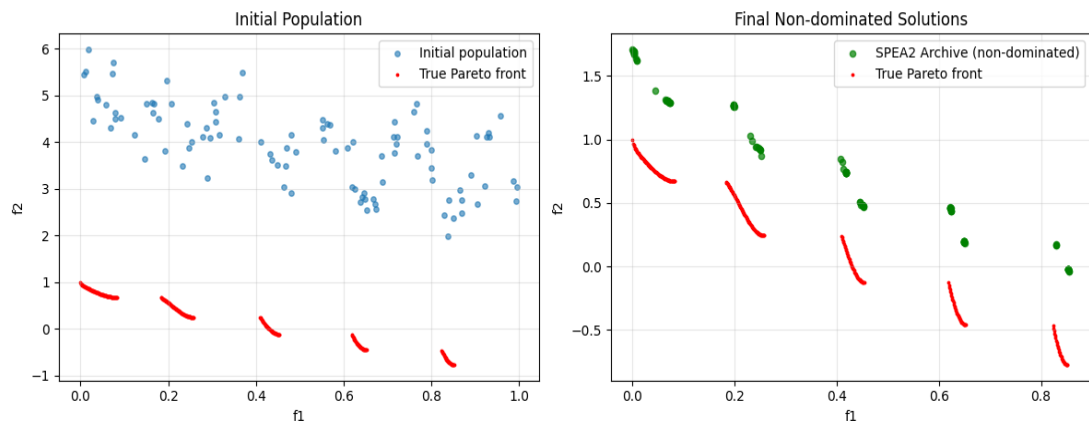


- **ZDT3 (Discontinuous)**

Generations: 200

Front-0 size at end : 60

SPEA2 Performance on ZDT3

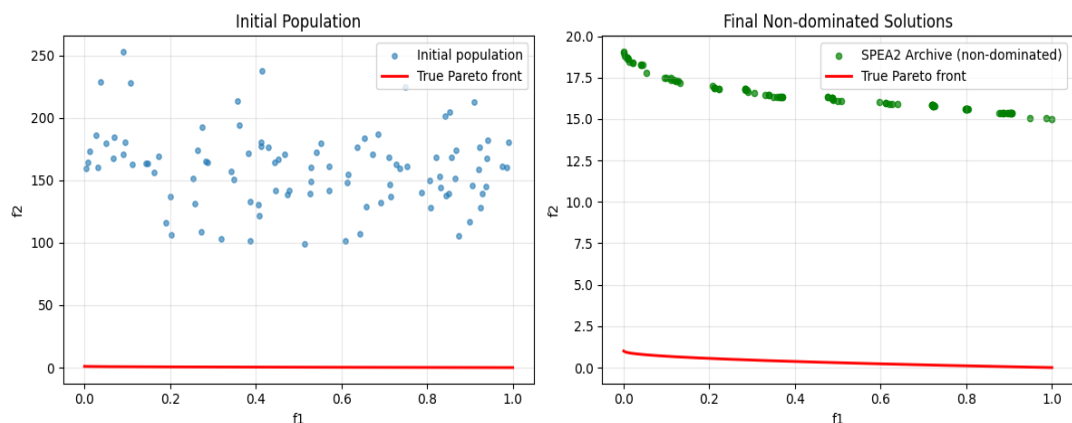


- **ZDT4 (Multi-modal)**

Generations: 400

Front-0 size at end : 400

SPEA2 Performance on ZDT4



- **ZDT1:** Both algorithms converge to the true Pareto front, which exhibits a convex nature. Although NSGA-II converges to a well-spread Pareto front, it lies considerably higher than the true Pareto front, implying that convergence has not been achieved yet. SPEA2 also fails to achieve convergence, but the solutions generated by this algorithm are a little bit clustered, especially in the mid region.
- **ZDT2:** The true Pareto front of ZDT2 is concave, and thus, it can be considered difficult to approximate. Again, convergence to the true Pareto front has not been achieved, and solutions lie far above the true Pareto front. The coverage of the range of f_1 is denser for NSGA-II compared to SPEA2.
- **ZDT3:** ZDT3 has a Pareto-optimal front which consists of five disjoint segments and thus it is the hardest problem among the four tested. Both algorithms correctly realize that the front has several segments and therefore their solution sets consist of several segments as well. Nevertheless, not all five segments are evenly covered by both algorithms – there are significantly more solutions in higher segments (f_1 is low) and no solutions in lower segments.

- **ZDT4:** It is believed that ZDT4 is the hardest problem among the tested because its objective function $g(x)$ is highly multimodal and thus it leads to 21^9 local Pareto optimal fronts. The initial population generated by both algorithms consists of very high values of f_2 (100-250), which indicates that this problem is deceptive. After 200 generations, both algorithms reach fronts significantly above the real Pareto-optimal front - NSGA-II solutions are situated in area $f_2 \in [19, 25]$ while SPEA2 solutions are located in the region $f_2 \in [15, 20]$.

6. Conclusion

The algorithms were applied and compared using four benchmark ZDT problems. NSGA-II and SPEA-II showed similar performance in terms of the overall results; that is, the correct identification of the shape of the Pareto frontier on problems ZDT1, ZDT2, and ZDT3, and poor results on problem ZDT4 because of its multi-modal nature.

On simple benchmark problems such as ZDT1 and ZDT2, NSGA-II gave slightly better convergence because of the better dispersion of solutions, which can be explained by the crowding distance selection used in this algorithm. For ZDT3, both algorithms managed to partially cover the discontinuous Pareto frontier without managing to completely cover any of the segments.

In general, neither of the algorithms was able to dominate the other. NSGA-II is easier and less computation-intensive because there is no need for the external archive, and it is thus a better choice for smooth, continuous Pareto fronts. The archive approach used by SPEA2 provides more protection of the non-dominated solutions and performs well in multimodal problems such as ZDT4. When deciding on which algorithm to choose for real-world applications, one must consider the problem's modality and Pareto front continuity.